

Schistosomiasis

- Schistosomiasis (**bilharziasis**) is a parasitic infection caused by **blood flukes (Trematodes)**.
- **Hepatic fibrosis**: responsible for most deaths
- **Chronic obstructive uropathy**: granulomatous inflammation
- Tx: **fresh water snails**: slow moving water of rivers, lakes and irrigation ditches



Epidemiology

- Only malaria accounts for more diseases than schistosomiasis
- WHO estimates
 - 200 million people are infected
 - 120 million display symptoms
 - 600 million people are at risk of infection
- 74 countries in Africa, Asia, and Latin America

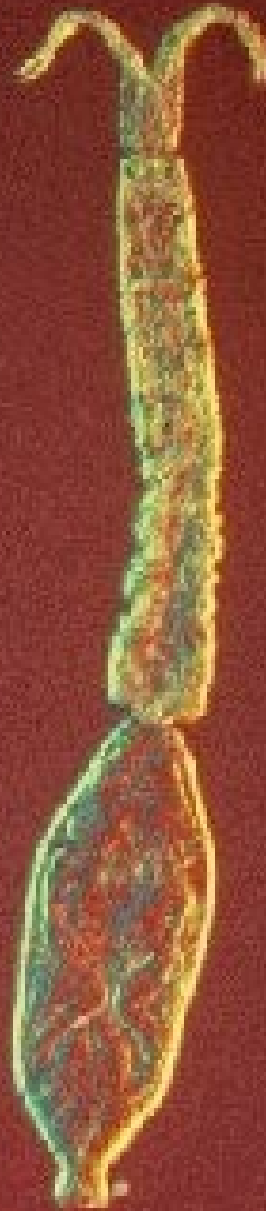
Parasite

- There are five species of schistosomes
 - *Schistosoma mansoni*
 - *S. haematobium*
 - *S. japonicum*
 - *S. mekongi*
 - *S. intercalatum*
- The 1st four are the **main** species that cause human infection

Life cycle

- The schistosome **egg hatches in water**,
- liberates a **miracidium** that penetrates a snail,
and
- develops through two stages to sporocyst to
form the final **infective larval stage**, the
cercaria

Cercaria: *infective larvae*

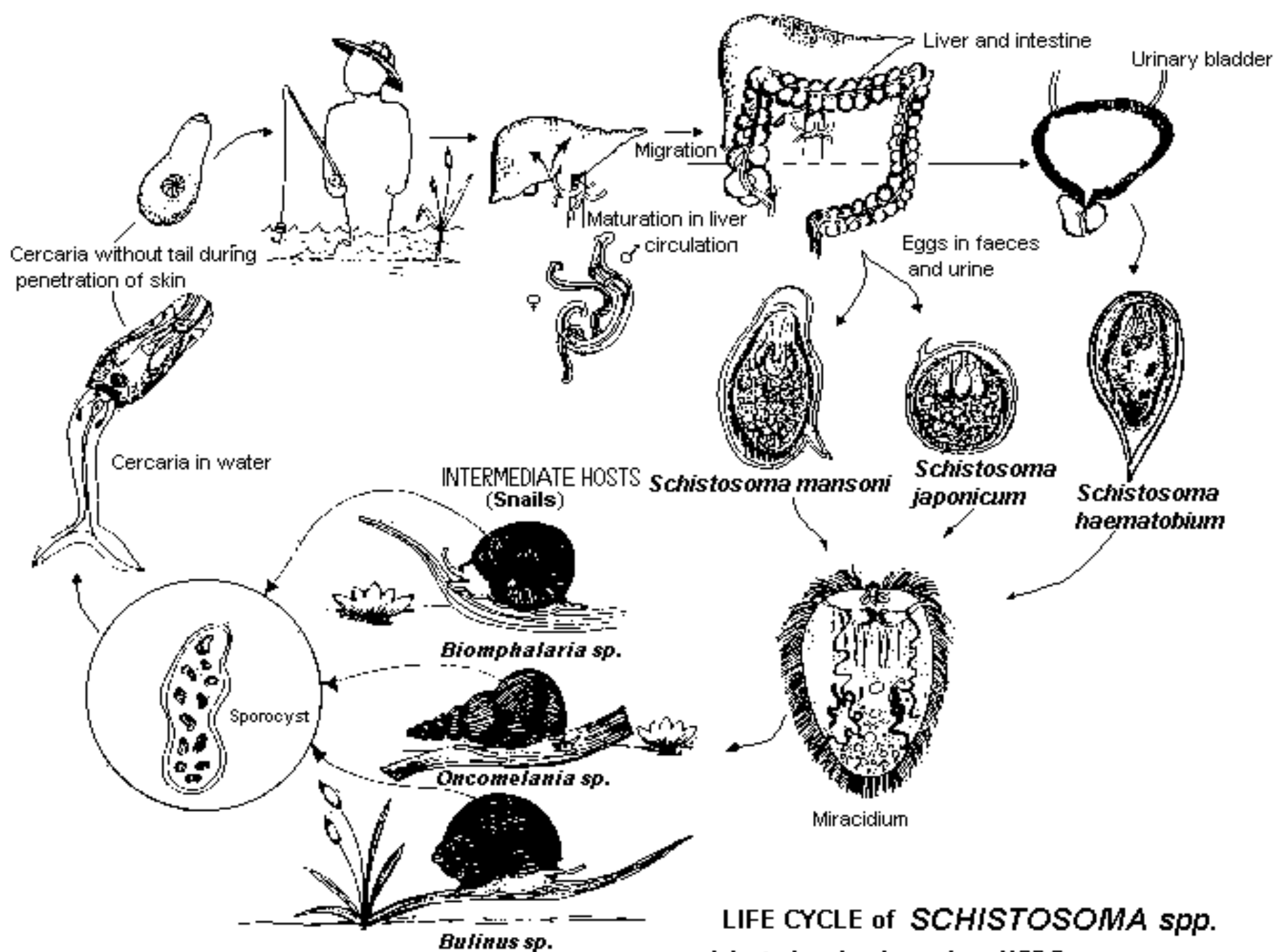


- The cercaria escapes from the snail into water,
-
- "swims" and penetrates the skin of a human host
- The cercaria loses its forked tail to become a schistosomule
-
- migrates through the tissues, penetrates peripheral blood vessel.

- Travel to the lung
- Matures and mate in hepatic vessels
- Migrate out as male-female worm pairs and settle in portal or pelvic venous system
- Female :hundreds of eggs per day => produce protease => elicit inflammation => passive transport of egg across intestinal and bladder wall => shed in stool and urine

- Infection of snails complete the life cycle
- Eggs taken to liver carried by portal circulation => fibrosing and granulomatous inflammation=> severe pathology of schistosomiasis.

- The female worm deposits immature eggs in small venules=> granulomas and fibrosis form around the eggs
 - of the intestine and rectum (S. mansoni and S. japonicum) or
 - of urinary bladder (S. haematobium).
- Man is the definite host harboring adult parasites, and fresh water snails are intermediate hosts



Pathogenesis

- The pathogenesis of *acute and chronic schistosomiasis* appears to involve immunologic mechanisms,
 - either humoral or
 - cell mediated.
- DTH due to secretion of soluble enzyme by the ova
 - Granuloma and fibrosis

S. mansoni

- found in
 - tropical Africa,
 - part of southwest Asia,
 - south America & Caribbean islands

S. mansoni

→ Clinical features (acute)

- Initially, at the site of entry of larvae, skin rash with pruritus, lasting for 1-2 days
- 3-8 weeks of symptom-free period
-
- appear allergic manifestations, known as Katayama Syndrome

Katayama Syndrome

- i) Fever, chills, sweating, headache & cough. ;
-
- ii) Urticaria, muscle & abdominal pain and patches of pneumonia ;
-
- iii) Splenomegaly

Chronic cases

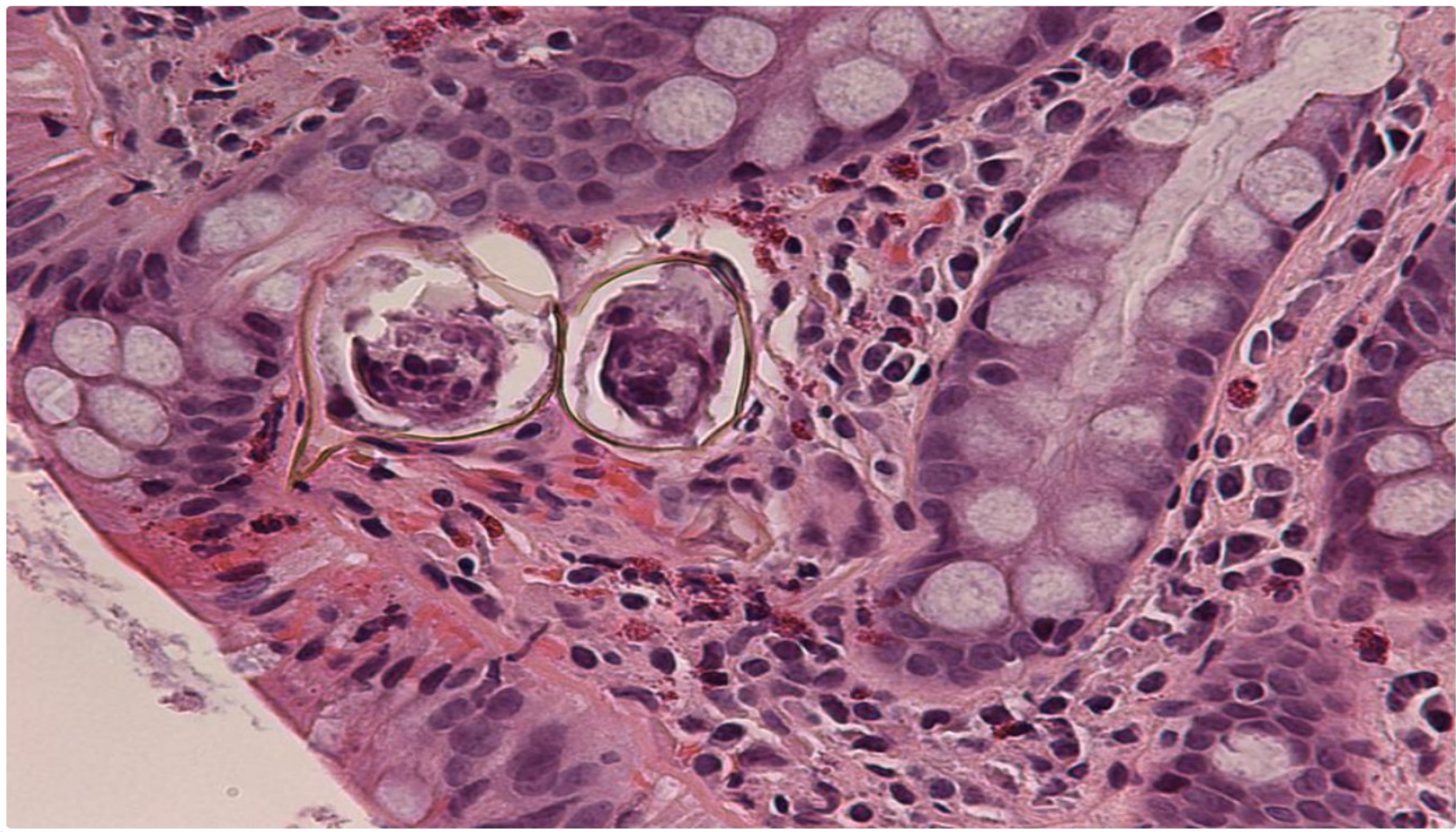
- i) **Portal hypertension** with esophageal varices & hemorrhoids causing hemorrhagic manifestations.(due to liver fibrosis)
- ii) **Pulmonary hypertension** causing cor-pulmonale
- iii) **Immune-complex Glomerulonephropathy**
- iv) **Transverse myelitis**- due to involvement of spinal cord

morphology

- Basic lesion is a circumscribed **granuloma**
-
- consisting of a centrally located round **ova** together with epithelioid cells,
- fibroblasts,
- giant cells and some plasma cells & **eosinophils**.

Intestinal Schistosomiasis

- Distal recto sigmoid colon is most commonly involved
- Ova are deposited in minute vessels of submucosa



Hepatosplenic Schistosomiasis

- Reticuloendothelial cell hyperplasia & granulomas are often seen in liver & spleen

Spleen

- Enlarged with fibrosis
-
- contain Gamma-Gandy bodies



S. japonica

- habitat of the organism is the lower mesenteric venous system
-
- In general S. Japonica is characterized by similar but more serious symptoms than S. Mansoni

S. hematobium

- Vesical and pelvic venous plexuses constitute the habitat



Clinical features

- i) Painless hematuria is the earliest symptom.
- ii) Other symptoms are, more or less, similar to *S. mansoni*.
- iii) Lesions in uterine cervix may simulate carcinoma of cervix.

Pathological changes in Urinary bladder

- Ova are deposited in **submucosa** of urinary bladder producing granulomas.
- Ova are excreted with urine

Pathological changes in Urinary bladder

- Mucosal changes →
 - hyperemia & petechiae
 - Polypoid or elevated
 - Glandular hyperplasia (cystitis cystica)
 - Squamous metaplasia
 - Ulceration

- Marked fibrosis around ureteric orifice causes
 - bladderneck obstruction with hydronephrosis and hydroureter
- In many patients carcinoma of the urinary bladder may develop.